EXPERIMENT 3: STUDY OF OPERATIONAL AMPLIFIER AS TRIANGULAR WAVE GENERATOR & ASTABLE MULTIVIBRATOR

OBJECTIVE:

- To study the basic functionality of an Operational Amplifier (Op-Amp).
- To analyze a bipolar triangular wave generator using Op-Amp.
- To analyze a unipolar triangular wave generator using Op-Amp.
- To design and test an astable multivibrator using Op-Amp.

INTRODUCTION:

Operational Amplifiers (Op-Amps) are versatile components extensively used in analog electronic circuits. They can be configured in a variety of ways to implement wave-shaping circuits, including triangular wave generators and multivibrators.

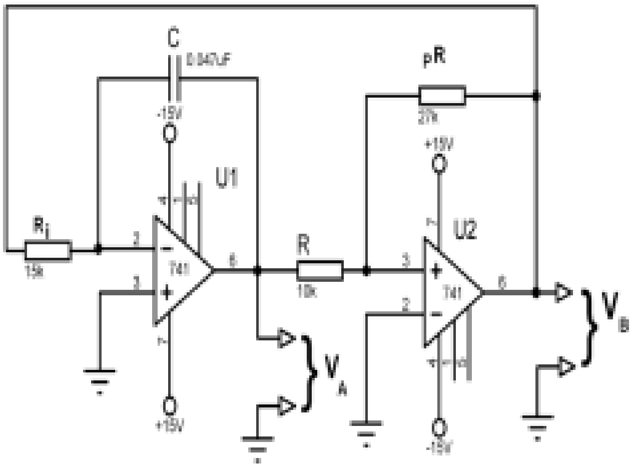
In this experiment, the Op-Amp is used to generate **bipolar** and **unipolar** triangle waves. These waveform generators are fundamental in signal processing and function generation. Additionally, the **astable multivibrator**—a circuit that oscillates continuously without external triggering—is also designed using Op-Amps. The output toggles between two voltage levels, generating a continuous stream of pulses, making it suitable for digital clock generation and timing applications.

The experiment emphasizes the utility of feedback networks and charging/discharging of capacitors in waveform generation, showcasing how Op-Amps form the core of many analog systems.

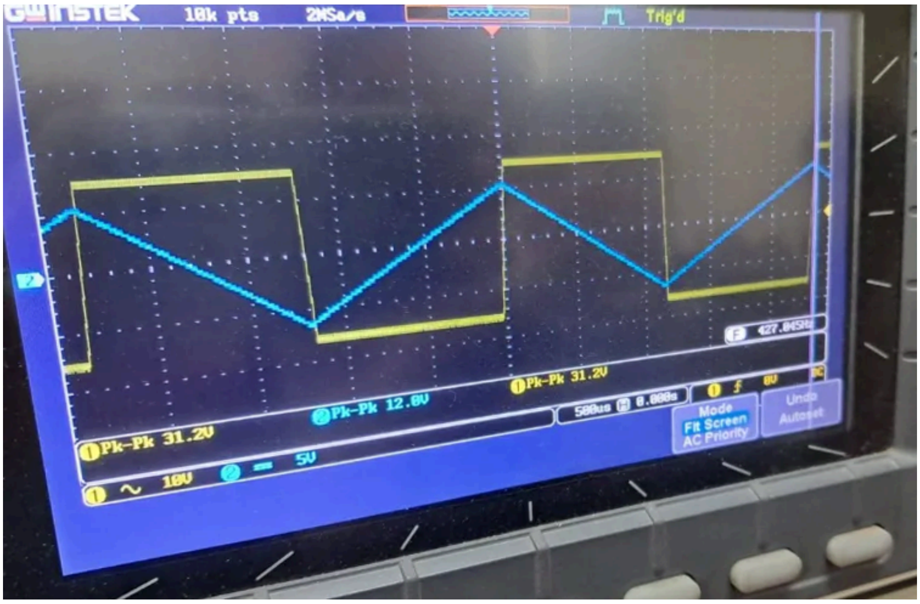
LIST OF EQUIPMENT:

- DC Power Supply
- Function / Signal Generator
- Breadboard
- Resistors
- Diode (IN4007)
- Capacitors
- Op-Amp (LM741)
- Oscilloscope
- Connecting wires

CIRCUIT DIAGRAM for Bipolar Triangle Wave Generator (Fig. 3c):



Graph:



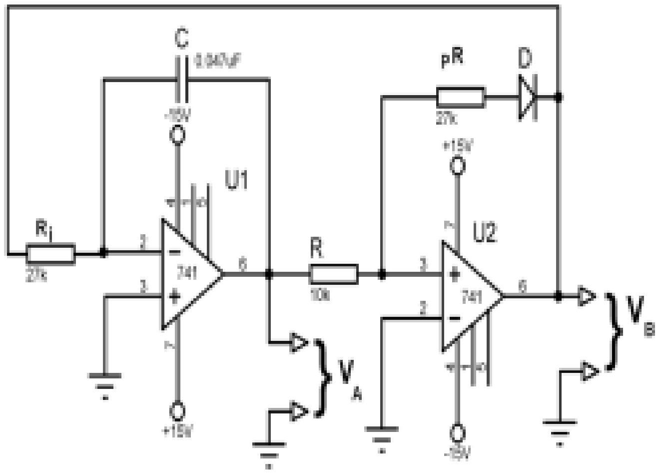
JUSTIFICATION / EXPLANATION OF THE EXPERIMENTAL DATA:

Concept:
This circuit uses an Op-Amp integrator and a square wave generator to produce a triangle wave that oscillates symmetrically around zero voltage (bipolar). The square wave alternates the current direction through the integrator, charging and discharging the capacitor, resulting in a triangle wave.

Experimental Result:

- The output waveform is triangular and centered at 0V.
- The frequency and amplitude depend on the RC values.
- The waveform confirms correct bipolar triangle generation.

CIRCUIT DIAGRAM for Unipolar Triangle Wave Generator (Fig. 3d):



Graph:

JUSTIFICATION / EXPLANATION OF THE EXPERIMENTAL DATA:

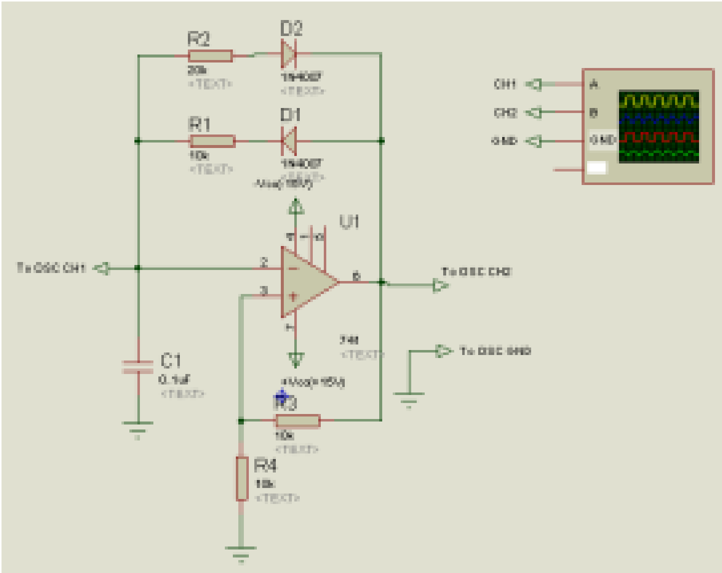
Concept:

Similar to the bipolar generator but biased such that the output remains in the positive voltage region (unipolar). Clamping and reference circuits may be used to set voltage levels appropriately.

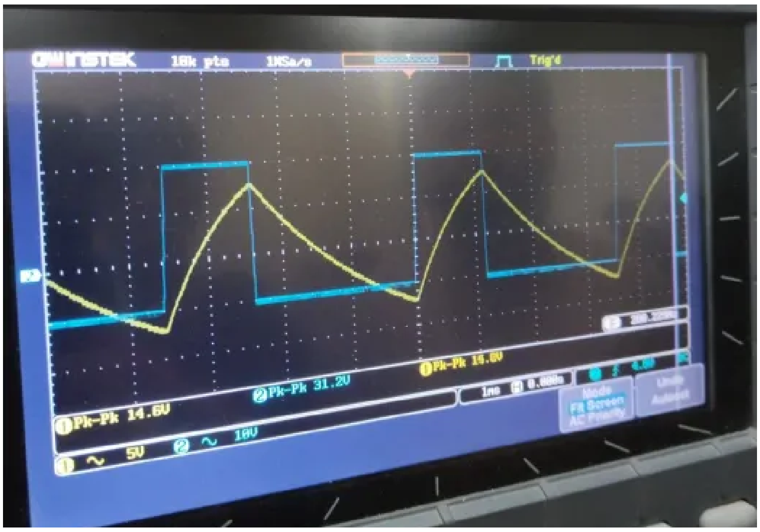
Experimental Result:

- Output is a triangle wave that varies between 0V and a positive voltage.
- Oscilloscope shows consistent periodic waveform.
- Output behavior matches expectations for unipolar triangle generator.

CIRCUIT DIAGRAM for Astable Multivibrator Using Op-Amp (Fig. 3e):



Graph:



JUSTIFICATION / EXPLANATION OF THE EXPERIMENTAL DATA:

Concept:
The circuit uses positive feedback through resistors and capacitors to form a regenerative loop that switches the Op-Amp output between high and low saturation levels, forming a free-running square wave oscillator.

Experimental Result:

- The output is a square wave without any external triggering.
- Frequency is determined by RC components.
- The waveform is symmetric with clean transitions.

ANSWERS TO REPORT QUESTIONS:

1. Analyze the differences between the circuits of Fig. 3(c) and Fig. 3(d).

- Biasing: Fig. 3(c) produces a bipolar waveform centered around 0V, while Fig. 3(d) is biased to keep the waveform in the positive voltage range.
- Output Nature: Bipolar triangle wave oscillates between positive and negative voltages; unipolar triangle wave remains above 0V.
- Component Differences: Unipolar configurations may include additional clamping or biasing elements.

---

2. Describe the circuit of Fig. 3(e) on the basis of its working principle.

- The astable multivibrator employs positive feedback to create oscillation without any external clock.
- A capacitor charges and discharges periodically through a resistor, and when threshold voltages are crossed, the Op-Amp switches output states.
- This behavior repeats indefinitely, generating a square wave at the output.

---

3. Write a short discussion on the conducted experiment.
In this experiment, the functionality of Op-Amps in waveform generation and multivibrator circuits was successfully demonstrated. The bipolar and unipolar triangle wave generators illustrated how integrators and feedback networks can be employed to synthesize precise waveform shapes. The astable multivibrator further expanded our understanding of Op-Amp applications in oscillators, showcasing how timing and feedback elements interact to generate square wave outputs. This experiment not only reinforced theoretical concepts of analog circuit design but also highlighted the versatility of Op-Amps in practical electronic systems.